

Steganography Analysis and Hidden Data Detection using StegExpose

EXPERIMENT NO: 08

DATE:

1. AIM

To conduct a comprehensive steganalysis lifecycle on digital image evidence using the StegExpose command-line utility; to evaluate Least Significant Bit (LSB) statistical properties and detection algorithms across a target evidence directory; to compute quantitative suspicion scores based on configurable thresholds; and to automatically compile, format, and export the analytical findings into a structured CSV report for forensic documentation.

2. SOFTWARE/TOOLS REQUIRED

- **Operating System:** Windows, Linux, or macOS
- **Runtime Environment:** Java Runtime Environment (JRE)
- **Forensics Tool:** StegExpose (Java jar executable)
- **Evidence Files:** Target image files (e.g., .png, .jpg, .bmp)

3. DESCRIPTION

StegExpose is a specialized steganalysis tool designed to detect the presence of hidden data within digital images. Unlike traditional steganography tools used to embed data, StegExpose serves a purely investigative role. It evaluates the statistical properties of an image to identify mathematical anomalies indicative of data hiding techniques, particularly Least Significant Bit (LSB) steganography.

The tool processes images and outputs a quantitative "suspect score" ranging from 0 to 1. By comparing this score against established thresholds, forensic investigators can

mathematically estimate the likelihood that an image contains clandestine data, allowing them to triage suspicious files rapidly without needing the original extraction password.

4. PROCEDURE

Step1: Open Command Prompt as Administrator

- Press the Windows Key, type cmd, right-click on **Command Prompt**, and select **Run as administrator**.

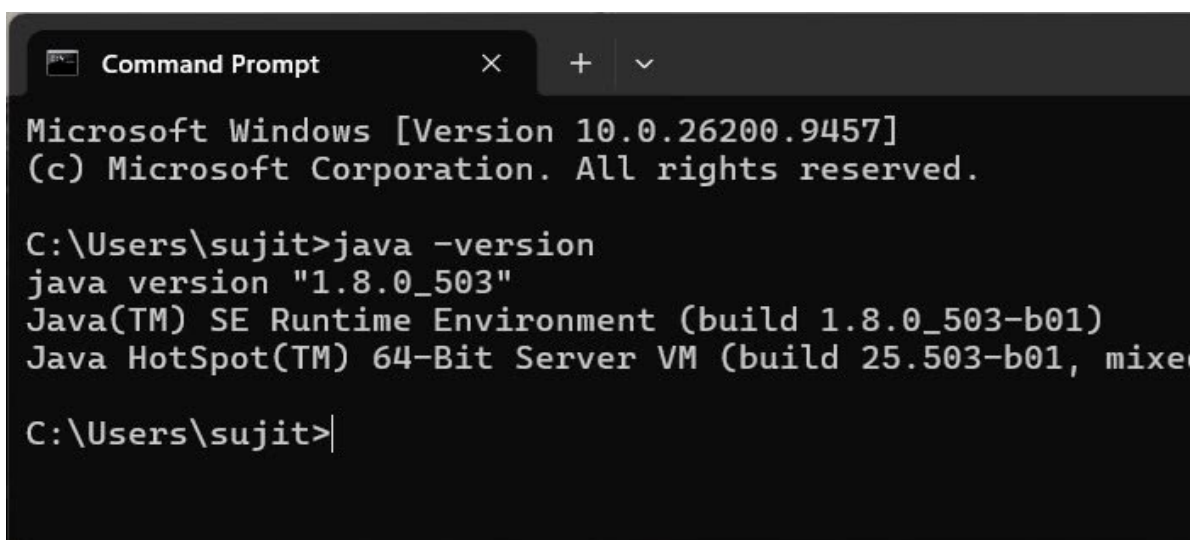
Step2: Navigate to the Working Directory

- Use the `cd` command to point the terminal to your project folder where `StegExpose.jar` and your evidence directory are located:
- And check if java is available.

DOS

```
cd /d "D:\Digital_Forensics_Lab\Experiment-08-Steganography\StegExpose-master"
java -version
```

(it shows if installed or not)

A screenshot of a Windows Command Prompt window. The title bar says 'Command Prompt'. The window content shows the following text: 'Microsoft Windows [Version 10.0.26200.9457] (c) Microsoft Corporation. All rights reserved. C:\Users\sujit>java -version java version "1.8.0_503" Java(TM) SE Runtime Environment (build 1.8.0_503-b01) Java HotSpot(TM) 64-Bit Server VM (build 25.503-b01, mixed mode) C:\Users\sujit>'.

```
Microsoft Windows [Version 10.0.26200.9457]
(c) Microsoft Corporation. All rights reserved.

C:\Users\sujit>java -version
java version "1.8.0_503"
Java(TM) SE Runtime Environment (build 1.8.0_503-b01)
Java HotSpot(TM) 64-Bit Server VM (build 25.503-b01, mixed mode)

C:\Users\sujit>
```

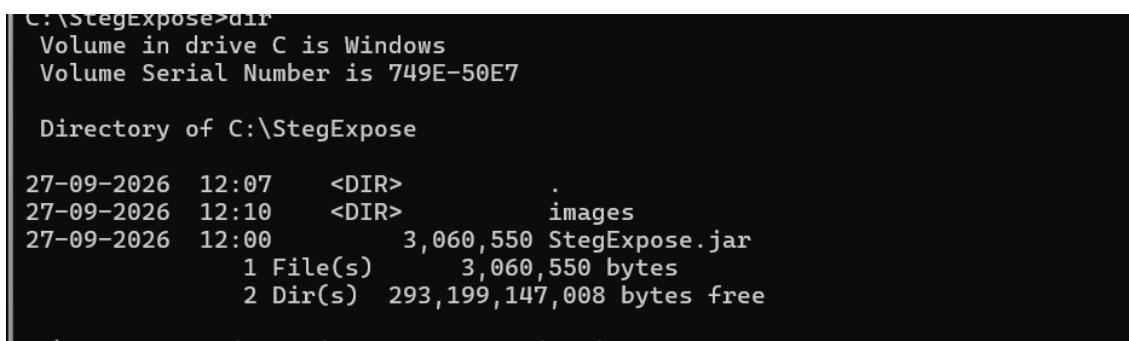
Figure 1: Navigating to the working directory in the Command Prompt

Step3:ExecuteStegExposeAnalysiswithCSVOutput

- Run the StegExpose command, passing the path to your evidence folder, the execution speed parameter (default), the statistical threshold (0.2), and the desired output report filename (report.csv):

DOS

```
java-jarStegExpose.jar"D:\Digital_Forensics_Lab\Experiment-08- Steganography\Evidence"
default 0.2 report.csv
```



```
C:\StegExpose>dir
Volume in drive C is Windows
Volume Serial Number is 749E-50E7

Directory of C:\StegExpose

27-09-2026  12:07    <DIR>          .
27-09-2026  12:10    <DIR>          images
27-09-2026  12:00             3,060,550 StegExpose.jar
               1 File(s)          3,060,550 bytes
               2 Dir(s)  293,199,147,008 bytes free
```

Figure2:ExecutingStegExposeviaCLIandredirectingoutputtoreport.csv

Step4:VerifyandReviewtheGeneratedReport

- Once the execution finishes, navigate to your folder or open the file in Microsoft Excel/Notepad to verify the generated report.csv contents.
- Otherwise,directlyopenthecsvfileon excel.

DOS

```
notepadreport.csv
```

	A	B	C	D	E	F	G	H	I	J	K	L	M
1													
2	File name	Above steg	Secret me	Primary Se	Chi Square	Sample Pa	RS analysi	Fusion (mean)					
3	test_image	FALSE	382	0.006294	0	null	0.0096	0.005298					
4													
5													
6													
7													
8													
9													
10													
11													
12													
13													
14													
15													
16													

Figure3: Opening and reviewing the generated CSV report detailing LSB scores and classifications

5. OBSERVATIONS

- **Batch Processing:** StegExpose scanned all image files within the target evidence folder sequentially.
- **Classification Metrics:** Each file was assigned an LSB score evaluated against the configured 0.2 threshold. Files exceeding the limit were flagged as suspected steganography carriers.
- **Report Compilation:** The console results were successfully written to report.csv, containing structured columns for filenames, scores, and status flags.

RUBRICS

Criteria&MarksAssigned	Mark Allotted	MarkAwarded
1.GitHubActivity& SubmissionRegularity	3	
2.Application of Forensic Tools&Practical Execution	3	
3.Documentation& Reporting	2	
4.Engagement,Problem-Solving&Team Collaboration	2	
Total	10	

6. RESULT

The steganography detection experiment was successfully executed using StegExpose. The tool scanned the evidence directory via the command line, analyzed combinations of LSB metrics, and generated a clean CSVreport detailing the cleanliness or suspicion level of each digital image artifact.